

modeldiag: A Unified Framework for Comprehensive Diagnostics of Statistical Models in R

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Summary

Statistical models underpin a broad range of applied research in medicine, ecology, economics, and the social sciences. Valid inference from these models depends on the satisfaction of structural assumptions—linearity, homoscedasticity, independence of errors, correct distributional form, and, in survival analysis, proportional hazards. When assumptions are violated, coefficient estimates may be biased, standard errors incorrect, and p -values misleading. The **modeldiag** R package (R Core Team 2024) provides a unified framework for diagnosing these assumption violations across four widely-used model classes: ordinary linear models, logistic regression, Poisson regression, and Cox proportional hazards models.

The package exposes a single generic function, `diagnose_model()`, which dispatches—via R’s S3 class system—to the appropriate battery of diagnostic checks for the detected model type. It returns an object of class `"model_diagnostics"` with `print()`, `summary()`, and `plot()` methods that deliver structured numerical results, human-readable interpretations, and publication-quality graphics respectively. Sixteen individual `check_*()` functions are also exported for targeted, reproducible diagnostic workflows.

Statement of need

Analysts who work across multiple model families—fitting linear models for continuous outcomes, logistic regression for binary outcomes, Poisson regression for counts, and Cox models for time-to-event data—must currently assemble bespoke diagnostic workflows from disparate packages. Relevant tests are spread across **car** (Fox and Weisberg 2019), **lmtest** (Zeileis and Hothorn 2002), **ResourceSelection** (Lele et al. 2019), and **survival** (Therneau and Grambsch 2000), each with its own argument conventions and output format. This fragmentation creates friction in reproducible analysis pipelines and demands that practitioners remember class-specific APIs.

modeldiag addresses this by offering a single, consistent interface that automatically selects and executes the right set of diagnostics for the fitted model class. The package targets practicing statisticians, epidemiologists, biostatisticians, and data scientists who need to validate model assumptions efficiently within a reproducible workflow.

State of the field

Several R packages address portions of the model-diagnostics problem. The **performance** package (Lüdtke et al. 2021) provides a `check_model()` function with visual output for linear, generalised linear, and mixed-effects models, but does not support Cox proportional hazards models or offer the full suite of GLM-specific diagnostics such as the Hosmer–Lemeshow goodness-of-fit test or zero-inflation detection. The **DHARMa** package (Hartig 2022) provides simulation-based quantile residuals for GLMs and mixed models; these residuals are powerful but require additional expertise to interpret and do not extend to Cox models. The **car** package (Fox and Weisberg 2019) provides generalised VIF computation and component-plus-residual plots but is not designed as a stand-alone, model-class-aware diagnostic workflow.

`modeldiag` was built rather than contributing to existing packages for the following reasons. First, no existing package provides a single `diagnose_model()` entry point that covers all four model classes (`lm`, `glm` with binomial and Poisson families, and `coxph`) with a fully consistent interface. Second, `modeldiag` couples test execution with structured text interpretation in its `summary()` method, making diagnostic output accessible to analysts who may not be familiar with the numerical output of individual test functions. Third, the package exposes both the high-level wrapper and the underlying `check_*()` functions, supporting both rapid exploratory diagnostics and granular, reproducible pipelines.

Software design

`modeldiag` implements an S3 dispatch system around a single generic, `diagnose_model()`, which branches to `diagnose_model.lm()`, `diagnose_model.glm()`, and `diagnose_model.coxph()` based on the class attribute of the fitted model. Each method executes the appropriate `check_*()` functions and returns a named list of class `"model_diagnostics"` containing the model type, the original model object, and all diagnostic results. The `print()`, `summary()`, and `plot()` S3 methods dispatch on this class to deliver formatted output and graphics.

The sixteen exported `check_*()` functions can also be called independently, enabling targeted diagnostics. For each model class the dispatched checks are:

- **Linear models (`lm`):** `check_vif()`, `check_heteroskedasticity()`, `check_autocorrelation()`, `check_linearity()`, `check_normality()`, `check_outliers()`
- **Logistic regression (`glm`, `binomial`):** `check_vif()`, `check_box_tidwell()`, `check_hosmer_lemeshow()`, `check_influential_glm()`, `check_separation()`
- **Poisson regression (`glm`, `poisson`):** `check_vif()`, `check_overdispersion()`, `check_zero_inflation()`, `check_influential_glm()`, `check_residual_analysis()`
- **Cox PH models (`coxph`):** `check_vif()`, `check_proportional_hazards()`, `check_influential_coxph()`, `check_functional_form_coxph()`

The package imports `car` (Fox and Weisberg 2019), `lmtest` (Zeileis and Hothorn 2002), `ResourceSelection` (Lele et al. 2019), and `survival` (Therneau and Grambsch 2000) for the underlying test computations, and depends on base R `stats` and `graphics` for all other operations.

Mathematics

Linear Models

The ordinary linear regression model is

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 I),$$

where $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$.

Multicollinearity. When predictors are nearly linearly dependent, the OLS estimator $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ has inflated variance. The Variance Inflation Factor for predictor X_j (Fox and Monette 1992) is

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination from regressing X_j on all other predictors. `modeldiag` classifies severity as negligible (< 2), moderate (2–5), high (5–10), or severe (≥ 10).

Heteroscedasticity. Homoscedasticity requires $\text{Var}(\varepsilon_i) = \sigma^2$. The Breusch–Pagan test (Breusch and Pagan 1979) regresses squared residuals on the original predictors; the test statistic is asymptotically χ^2 under $H_0: \text{Var}(\varepsilon_i) = \sigma^2$.

Autocorrelation. Independence of errors requires $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. The Durbin–Watson statistic (Durbin and Watson 1950, 1951) is

$$DW = \frac{\sum_{i=2}^n (\hat{\varepsilon}_i - \hat{\varepsilon}_{i-1})^2}{\sum_{i=1}^n \hat{\varepsilon}_i^2},$$

with values near 2 indicating no autocorrelation.

Linearity. The Ramsey RESET test (Ramsey 1969) augments the model with powers of the fitted values $\hat{Y}^2, \hat{Y}^3, \dots$ and tests their joint significance. Rejection indicates that $\mathbb{E}[Y \mid X]$ may not be correctly specified as linear in X .

Normality of errors. The Shapiro–Wilk test (Shapiro and Wilk 1965) evaluates $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ via

$$W = \frac{(\sum_{i=1}^n a_i \hat{\varepsilon}_{(i)})^2}{\sum_{i=1}^n (\hat{\varepsilon}_i - \bar{\varepsilon})^2},$$

where $\hat{\varepsilon}_{(i)}$ are order statistics and $\mathbf{a}$ derives from the expected values of normal order statistics.

Influential observations. Cook’s distance (Cook 1977; Cook and Weisberg 1982) measures the aggregate shift in $\hat{\beta}$ upon deletion of observation i :

$$D_i = \frac{(\hat{\beta} - \hat{\beta}_{(i)})^\top X^\top X (\hat{\beta} - \hat{\beta}_{(i)})}{p \hat{\sigma}^2}. \quad (1)$$

Observations with $D_i > 4/n$ (see Equation 1) are flagged as potentially influential.

Logistic Regression

Logistic regression models a binary outcome through

$$\text{logit}(P(Y = 1 \mid X)) = \log\left(\frac{p}{1-p}\right) = X\beta.$$

Linearity of the logit. The Box–Tidwell test (Box and Tidwell 1962) augments the model with interaction terms $X_j \ln(X_j)$ and tests whether their coefficients are zero, evaluating whether each continuous predictor enters the log-odds linearly.

Goodness of fit. The Hosmer–Lemeshow test (Hosmer et al. 2013) partitions observations into $g = \min(10, \lfloor n/5 \rfloor)$ groups based on estimated probabilities and computes

$$\hat{C} = \sum_{k=1}^g \frac{(O_k - E_k)^2}{E_k(1 - E_k/n_k)},$$

where O_k and E_k are observed and expected event counts in group k . Under good fit, $\hat{C} \sim \chi_{g-2}^2$.

Separation. Complete or quasi-complete separation occurs when a linear combination of predictors perfectly classifies the outcome, causing maximum-likelihood estimates to diverge. `modeldiag` detects this by inspecting coefficient values and model convergence status.

Poisson Regression

Poisson regression assumes $Y \mid X \sim \text{Poisson}(\lambda)$ with $\log(\lambda) = X\beta$ and the equidispersion property $\text{Var}(Y) = \mathbb{E}(Y) = \lambda$.

Overdispersion. When $\text{Var}(Y) > \mathbb{E}(Y)$, standard errors are underestimated. The ratio (Dean and Lawless 1989; Cameron and Trivedi 2013)

$$\phi = \frac{D_{\text{resid}}}{\text{df}_{\text{resid}}}$$

provides a practical diagnostic; `modeldiag` flags $\phi > 1.5$ as evidence of overdispersion.

Zero inflation. When the observed proportion of zeros substantially exceeds the Poisson expectation,

$$\hat{p}_0^{\text{obs}} > 1.5 \hat{p}_0^{\text{exp}}, \quad \hat{p}_0^{\text{exp}} = \frac{1}{n} \sum_{i=1}^n e^{-\hat{\lambda}_i},$$

`modeldiag` flags a potential zero-inflated process, prompting consideration of ZIP or ZINB models.

Cox Proportional Hazards Model

The Cox model (Cox 1972) specifies the hazard as

$$h(t \mid X_i) = h_0(t) \exp(X_i\beta),$$

where $h_0(t)$ is an unspecified baseline hazard and $\exp(\beta_j)$ is the hazard ratio for a one-unit increase in X_j .

Proportional hazards. The model assumes the hazard ratio is constant over time. Schoenfeld residuals (Schoenfeld 1982) are uncorrelated with time under H_0 ; the Grambsch and Therneau (1994) test formally evaluates

$$H_0: \frac{\partial \beta(t)}{\partial t} = 0$$

via a weighted correlation between scaled Schoenfeld residuals and a monotone function of time, implemented through `survival::cox.zph()`.

Influential observations. Dfbeta residuals approximate the change in $\hat{\beta}_j$ upon deletion of observation i . Observations with $|\text{dfbeta}_{ij}| > 0.2$ for any coefficient are flagged as influential.

Functional form. Martingale residuals—the difference between the observed event indicator and the expected cumulative hazard—are plotted against continuous predictors to detect non-linear relationships not captured by $X\beta$.

Usage

Linear Models

The full diagnostic suite for a linear model is obtained with a single call to `diagnose_model()`. The `summary()` method prints each result alongside a plain-language interpretation.

```
library(modeldiag)

model_lm <- lm(mpg ~ wt + hp + disp, data = mtcars)
diag_lm <- diagnose_model(model_lm)
summary(diag_lm)
```

Individual checks are also available for targeted analysis:

```
check_vif(model_lm)
check_heteroskedasticity(model_lm)
check_autocorrelation(model_lm)
check_linearity(model_lm)
check_normality(model_lm)
check_outliers(model_lm)
```

The `plot()` method renders a grid of diagnostic panels including residuals versus fitted values, a normal Q-Q plot, scale-location, residuals versus leverage, and the autocorrelation function of residuals (Figure 1).

```
plot(diag_lm)
```

Logistic Regression

```
model_glm <- glm(am ~ wt + hp, data = mtcars, family = binomial)
diag_glm <- diagnose_model(model_glm)
summary(diag_glm)
```

Individual logistic regression diagnostics:

```
check_vif(model_glm)
check_box_tidwell(model_glm)
check_hosmer_lemeshow(model_glm)
check_influential_glm(model_glm)
check_separation(model_glm)
```

Poisson Regression

```
model_pois <- glm(carb ~ wt + hp, data = mtcars, family = poisson)
diag_pois <- diagnose_model(model_pois)
summary(diag_pois)
```

Individual Poisson diagnostics:

```
check_vif(model_pois)
check_overdispersion(model_pois)
check_zero_inflation(model_pois)
check_residual_analysis(model_pois)
```

Cox Proportional Hazards Model

```
library(survival)
data(lung)

model_cox <- coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
diag_cox <- diagnose_model(model_cox)
summary(diag_cox)
```

Individual Cox model diagnostics:

```
check_proportional_hazards(model_cox)
check_influential_coxph(model_cox)
check_functional_form_coxph(model_cox)
```

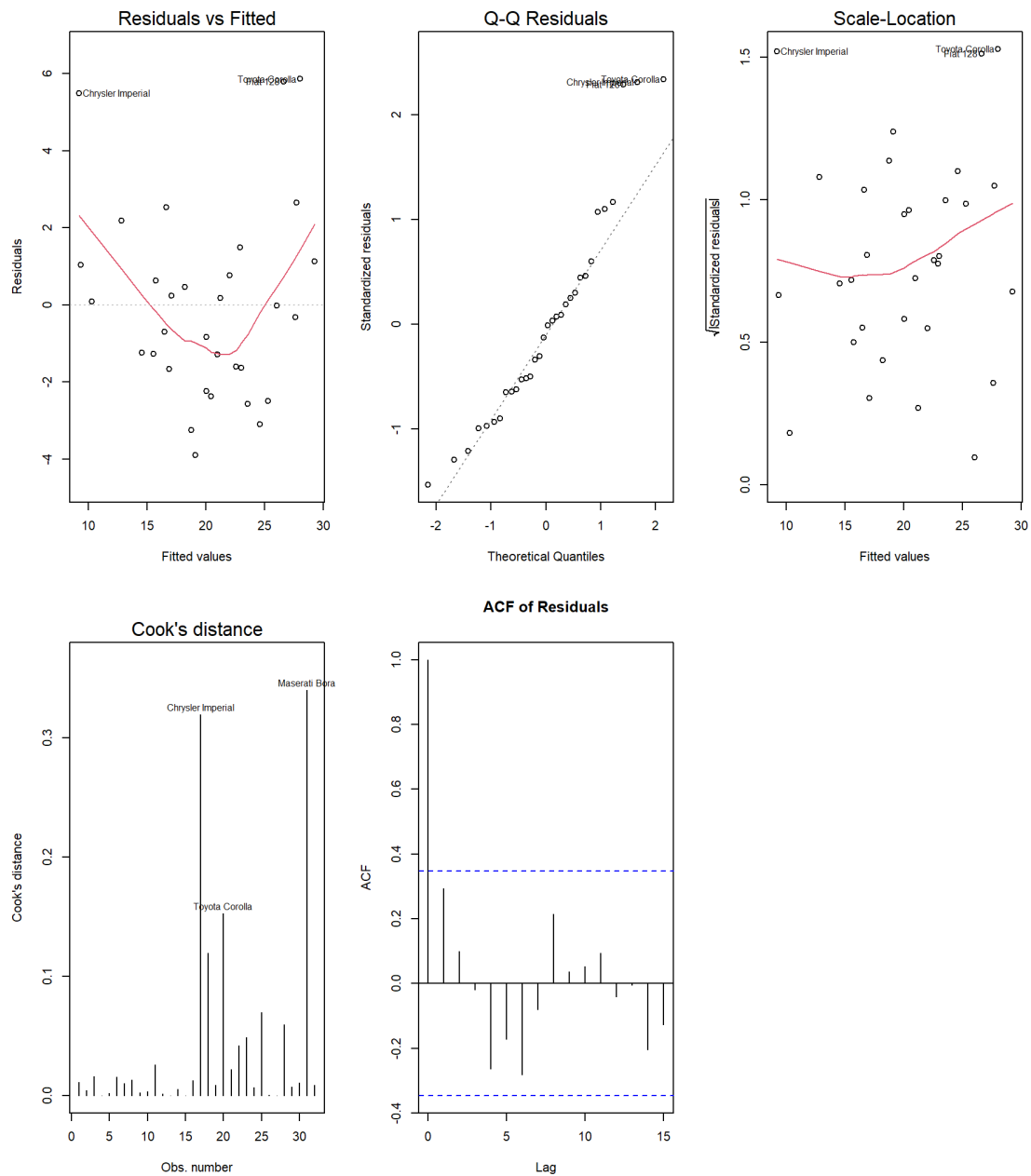


Figure 1: Diagnostic plots for the linear model fitted to the mtcars dataset.

Diagnostic plots for the Cox model (Figure 2) display Schoenfeld residual trends over time, dfbeta influence plots for each coefficient, and martingale residuals against continuous predictors.

```
plot(diag_cox)
```

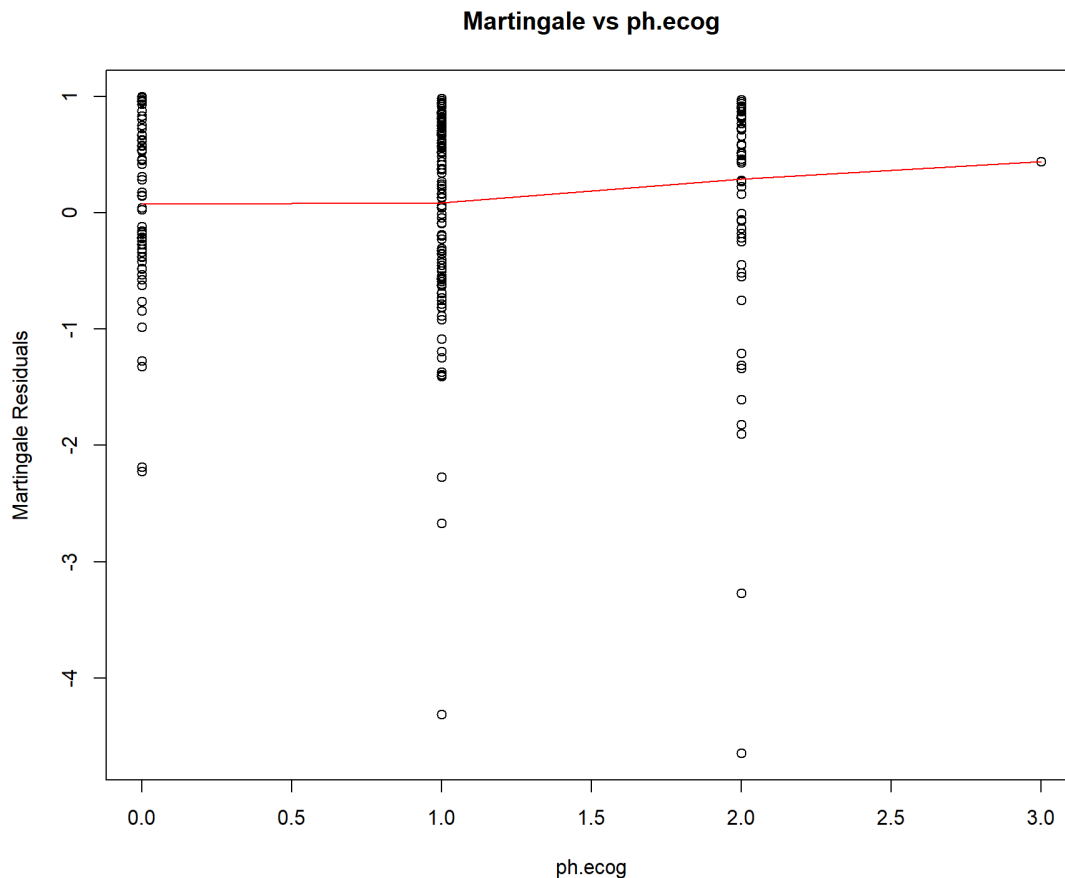


Figure 2: Diagnostic plots for the Cox proportional hazards model fitted to the lung dataset.

AI usage disclosure

No generative AI tools were used in the development of this software or the writing of this manuscript.

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